Chapter 4 Food Preservation Methods

Learning Objectives

This chapter will enable the students to

- Know the importance of food preservation
- Enhance their knowledge of different preservation techniques
- Know the difference between traditional techniques and modern industrial techniques
- Understand the steps to be taken before freezing vegetables, fruits, meat and poultry
- Know the different methods of food preservation
  - *Using low and high temperature*
  - *Use of chemical preservatives and high osmotic pressure*
  - *Dehydration*
  - *Irradiation*
  - *Vacuum packaging*

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4.1 INTRODUCTION

Millions of fruits and vegetables are produced each year and they are lost due to poor processing and preservation. Fresh fruits are abundant during the season and are not available during off season. Due to this the food has to be stored until the next season. Fish and meat too have to be preserved as all that is killed or caught cannot be eaten at one time. Bacteria, fungi and yeasts tend to decay the food and render it unfit to eat. Hence all fresh foods have to be preserved if it is to be used after a period of time.

Besides when food spoils, they undergo physical and chemical changes that results in the food becoming inedible or hazardous to eat. The chief *causes of food spoilage* are;

- *The growth of microorganisms like bacteria, yeast and moulds.*
- *The action of enzymes that normally occur in the food.*
- *Other causes of spoilage are non enzymatic reactions in food such as oxidation and mechanical damage such as bruising and damage from rodents and insects.*

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